



Joint inversion of DC/TDEM data: a case study of static shift problem in the area north Tenth of Ramadan City, Egypt

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Received: 5 July 2020 / Accepted: 16 June 2021 / Published online: 29 July 2021
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Abstract

Discrete resistivity (DC) and time-domain electromagnetic (TDEM) techniques have shown their effectiveness for studying shallow subsurface features including groundwater. The main objective of this research is to discuss the static shift problem of DC data that is noticed in the study area, north Tenth of Ramadan City, Egypt. The DC and TDEM joint inversion help to solve such problem of static shift due to near surface inhomogeneities. Therefore, thirty-two soundings of each of DC and transient electromagnetic TDEM methods were jointly conducted in the study area. A scaling factor that ranges between 1.14 and 1.5 has been applied to overcome the static shift of DC data. The output results showed five geoelectrical layers of Quaternary rock units varied laterally and vertically in their thickness and lithological composition from fine sands and gravels with clay intercalations. The fifth layer may represent the water saturated sand representing unconfined aquifer as it has resistivity ranging from 3.5 Ω .m to 21.7 Ω .m with a depth ranging between 51 m and 139 m. This depth is also consistent with the hydrogeological background of the study area (ranges from 60 to 80 m). The resistivity values generally decrease towards the north because of surface water infiltrations from Ismailia irrigations Canal. The southern part of the study area is affected by two normal faults that greatly control upward water leakage and interconnection between shallow and deep quaternary aquifers.

Keywords Electrical resistivity · Transient electromagnetic · Vertical electrical sounding · Joint inversion · Static shift · Groundwater · Nile Delta

Introduction

Groundwater is a valuable freshwater resource constituting about two-third of the freshwater reserves of the world (Bayowa et al. 2018; Ren et al. 2021; Wang et al. 2020). The good quality of groundwater, its accessibility and its relatively low cost are factors that stimulate an increasing level of the exploitation of this resource (Abo-El-Fadl 2013). The lack of fresh water is one of the primary world issues that has affected many countries in the arid and semi-arid area (Li

et al. 2018, 2019; Wu et al. 2020). Groundwater depletion is one of the most common tragedies in these areas, causing many villages and cities to be abandoned (Hashemi et al. 2019). Also, lack of surface water due to climatic change and construction of dams on the water sources in the riparian countries increases the importance of groundwater as an alternative source (Hussain and Abed 2019). Therefore, it is requiring knowledge and study on groundwater at the above areas in order to control and organize its usage, as well as its quality and suitability for domestic, irrigation and industrial purposes (Kalhor et al. 2019; Hussain and Abed 2019). The eastern part of the Nile Delta recently became one of the most promising areas for development of land reclamation for agriculture, new residential areas, and industrial development. Accordingly, our need for water resources to meet the domestic, irrigational, and industrial demands has increased. Unfortunately, the available hydrological information and water wells in this area cannot cope with the needed requirements, as we have only one water well P-2 at the western border of the studied area (Fig. 1b). Therefore, the need for

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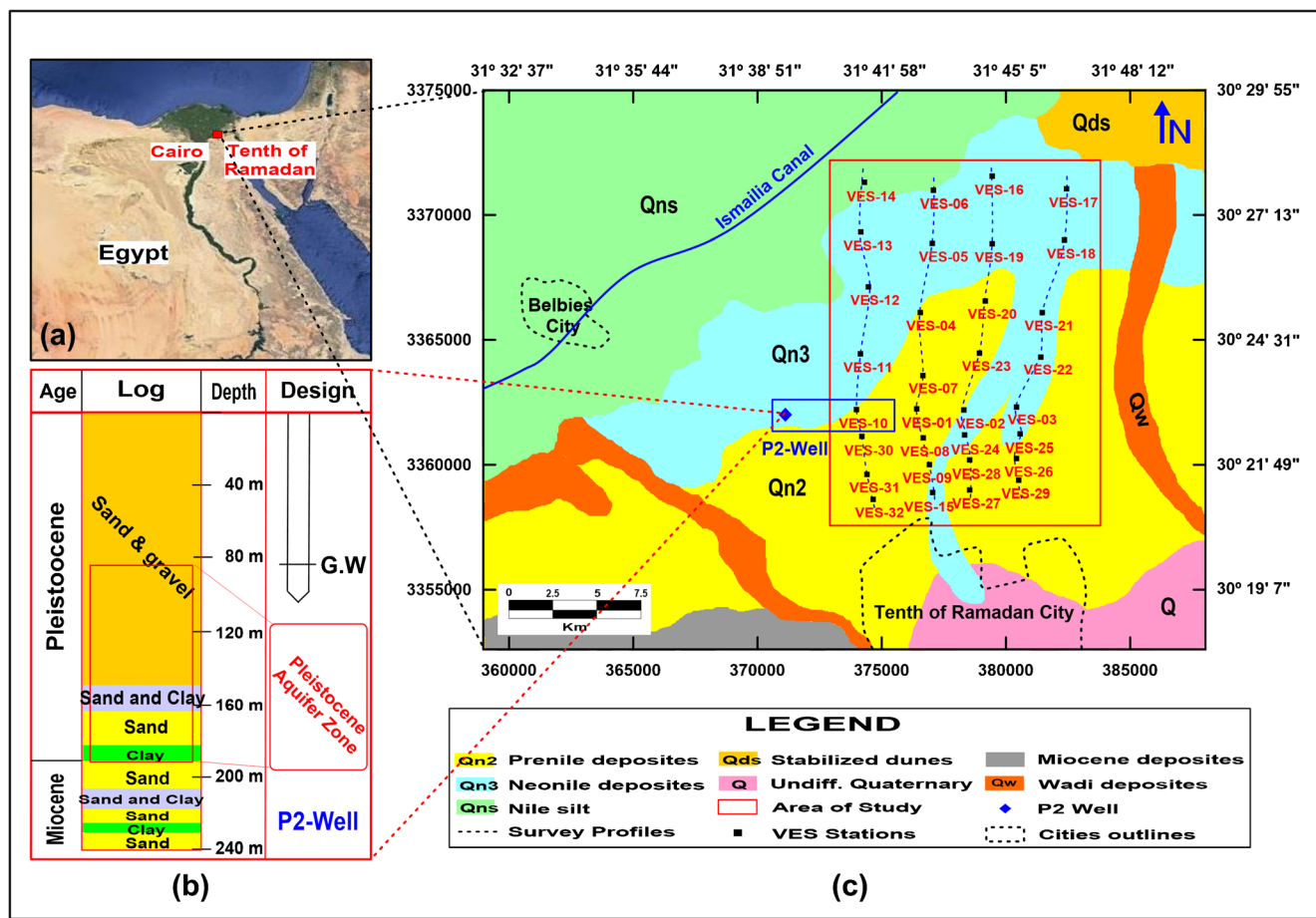


Fig. 1 (a) General location map of study area eastern Nile Delta fringe, (b) Lithologic log of Well-P2 west of the study area, and (c) surface geological map of the study area (Conco 1987)

groundwater investigation has become inevitable through geoelectrical surveys to detect and map the shallow free Quaternary aquifers existing in the study area.

The Discrete resistivity (DC) and time-domain electromagnetic (TDEM) geophysical methods are of most importance for studying shallow subsurface geological features and groundwater aquifers. This is due to its usefulness to provide valuable information about the subsurface lateral variations in lithology and structures at reasonable depths (Daniel et al. 2019). The interpretation of geophysical data is suffering from the influence of equivalence where there exists a set of models, which all equally well satisfy the data response. So, the different geophysical methods in a complementary form can be jointly used to reduce the ambiguities and get more trustful model (Daniel et al. 2019). Many authors studied the soil characterizations and groundwater investigation in the study area by using DC electrical resistivity method (El-Mahmoudi et al. 2006; Eleraki et al. 2010; Attwa et al. 2013, 2014; Massoud et al. 2016). On the other hand, Shaaban et al. (2016) discussed shallow groundwater investigation using TDEM method at Itay El-Baroud, western Nile Delta. The phenomenon of static shift problem in the area was

not recorded. Also, Sultan and Santos (2009) applied the TDEM/resistivity joint inversion for groundwater exploration in the northern eastern part of Greater Cairo.

The main objectives of this research are to investigate the shallow Quaternary-Pleistocene rock units and expected unconfined aquifer using transient electromagnetic TDEM and DC resistivity methods and to assess the joint inversion techniques between TDEM and DC data, and how to overcome the static shift problem which are clearly noticed in this work to enhance our interpretation result. The results could help to enhance our knowledge about shallow subsurface geology and existing shallow aquifers in the area and consequently suggesting sites for further drilling water wells for reclamation investment.

The study area

The concerned study area is located on the fringes of the southeastern Nile Delta region, Egypt. It is bounded by Belbies City and Ismailia Canal in the north, Tenth of Ramadan City in the south, Cairo-Belbeis desert road in the

west, and El Shabab Canal in the east (Fig. 1c). It is located between latitudes $30^{\circ} 21'$ and $30^{\circ} 29'$ and longitudes $31^{\circ} 41'$ and $31^{\circ} 47'$, covering an area of about 109 km^2 . Topographically, it is mostly arid area ranging in elevation from 25.5 m amsl in the northwest and 99.5 m amsl in the southeast with a general slope towards the north.

Climate

The study area is characterized by desert belt of Egypt as the arid conditions and weakness of chemical weathering are prevailed during most of years. So, the climate of the area is hot and rainless in summer and moderate cold, moist with little amount of precipitation in winter. Wind action usually forms desert pavement in soils of deltaic stage and accumulates sand dunes in plains. The main factor affecting the differentiation of soil properties may be attributed to the relief presence (Gobran et al. 1992).

Geological setting

Geomorphologically, the area of study is mainly characterized by Tenth of Ramadan- Al Salhiya gravelly and sandy plains. It is bounded by Wadi El Tumilat depression in the north. Geologically, it has a sedimentary succession ranging in age from Cretaceous to Quaternary sediments (Fig. 1b,c). The Cretaceous deposits consist of shales and sands in the lower part with thickness up to 355 m and dolomitic and marly limestones in the upper part with 310 m thick (Taha et al. 2003). The Quaternary deposits are composed of eolian sands, fluvial deposits, gravels and clays (Fig. 1c) covering a large part (85%) of the study area, particularly to the north and northeast (Conco 1987). The remaining part of the study area (15%) is covered by Tertiary sands and clays of shallow marine origin (Said 1981). Structurally, faults of mobile belt zone detected in the study area are of a normal type, mainly striking in E-W and NW-SE which were initiated during the Late Oligocene time. A mobile belt zone characterizes this area (Said 1990).

Hydrogeological setting

Tenth of Ramadan City and its surroundings are characterized by the main unconfined Quaternary freshwater aquifer (Fig. 2). The depth to the groundwater varies from 60 m to 90 m (RIGW 2005). The groundwater salinity in the investigation area varies between 750 and 950 mg/l (Ibrahim and Nofal 2020). The aquifer is recharged mainly from three sources; the seepage from the Nile Delta aquifer, the seepage from the fresh water of Ismailia canal, and the seepage from agricultural water uses (Atwa et al. 2015). On the other hand, the deeper Miocene aquifer is dominated by clastic facies in the southern part of

Tenth of Ramadan City and is overlain by about 200 m thick of Quaternary deposits (Khalaf and Gad 2015).

The understanding of the two aquifers configuration and confining clay units shows neither of the two aquifers is continuous for the entire length of the study area. Both aquifers are interconnected in the southern central part of the study area where the groundwater is mixed (Gad et al. 2015; Khalaf and Gad 2015). In the study area, the flow of the groundwater is local in both aquifers (Ismail 2008). The thickness of the clay layer and its classification are very important factors for the water infiltration and the vulnerability of underlying aquifers to pollution (EEBS 2004; Christiansen et al. 2006). So, the study area is located within a medium to high groundwater vulnerability (Khater 2002). This is because thick clay cap rocks are mostly absent or very thin in this area. Moreover, the probable higher levels of chloride in the ground water in desert tract is attributed probably to potential causes of contamination (Solangi et al. 2019) or to movement of the groundwater through regions where impurities will be dissolved and deteriorating its quality (Saba et al. 2016; Sridharan and Nathan 2017). In the vicinity of the Ismailia Canal (which runs through unit B), the aquifer is directly connected with the Canal, and the wells located nearby the Ismailia Canal show a depth to water to be less than 7 m (Al Shahat et al. 2014). The bottom of the Canal cuts through the sands and gravels of the Quaternary deposits. It allows the free seepage of surface water down to groundwater which is attributed to the presence of coarse sands in the banks and the bottom of the Ismailia Canal and the coarser facies of unit B (Abdulaziz 2007).

Methodology

Data acquisition and processing

The geo-electrical data acquisition was carried out in February (2017) using DC and TDEM surveys. The spacings between these sites are controlled by the accessibility of the terrain. Vertical electrical soundings (VES) of the DC resistivity method were measured by the Syscal/R2 acquisition system at 32 stations using the Schlumberger electrodes configuration. The current electrode separation (AB) began with 2 m and extended to reach a maximum distance of 1000 m in order to access to the depth range of shallow groundwater aquifers that fulfill the aim of the study.

On the other hand, TDEM soundings were collected using SIROTEM MK3 instrument to detect underground conducting materials by transmitting electrical pulses ($>5\text{A}$) along a single loop configuration ($50 \times 50 \text{ m}$) of a cable laid out on the ground surface. In the initial meters of top surface layer, TDEM signal has no resolution but it shows the layer

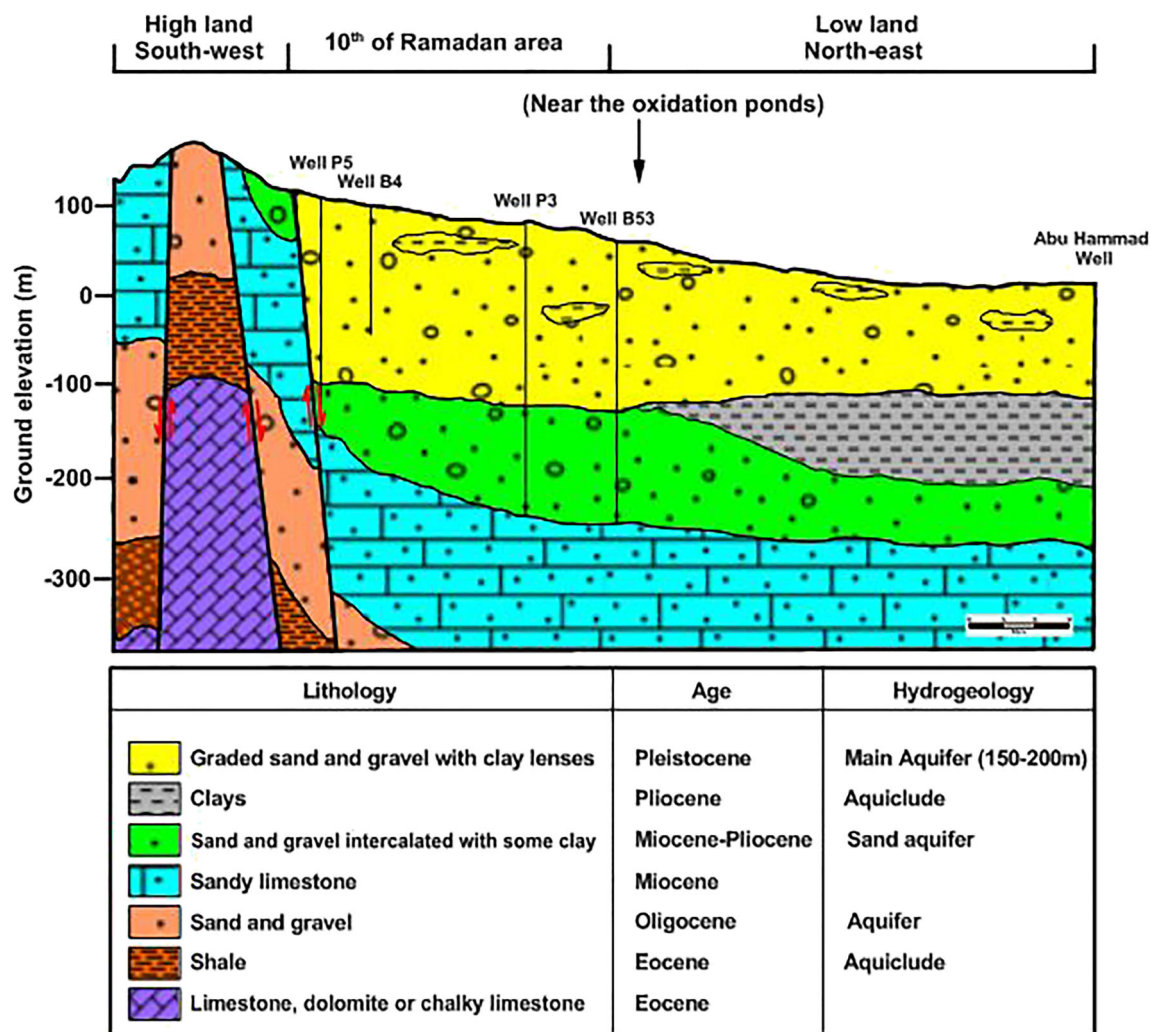


Fig. 2 Hydrogeological cross-section (A–B) north of the Tenth of Ramadan City and its surroundings (El-Fayoumy 1968; EEBS 2004)

composition which varies between soil, transition zone and portion of saturated sediments where average resistivity of these compositions is calculated (Leite et al. 2018). A separate modeling of DC and TDEM data might give discrete output results such as resistivity, thickness, and depths, because they investigate the subsurface in different ways (Raiche et al. 1985). This process carries ambiguity (non-uniqueness) phenomenon in interpretation with poorly determined geophysical problems.

According to Bortolozzo et al. (2017) and Leite et al. (2018), the joint inversion technique between DC and TDEM data was applied to remove the static shift phenomenon and to make integration between the results of the two methods. The shallow resistive layers are defined better by DC method while the deeper layers and conductive structures are well identified by TDEM method (Leite et al. 2018). When DC data is jointly inverted with TDEM data, a solution space which satisfies both data sets is reduced and the parameter estimation is more reliable (Leite et al. 2018).

Static shift between DC and TDEM data

When there is a 2D conductive structure at shallow depth (Fig. 3a), the propagated EM waves cause an accumulation of charges on the body surface resulting in the distortion of the electric field, in which there is either a concentration or sparse deployment of current density around the anomalous body. This may lead to an increase or decrease in the electric field component measured at the earth's surface and hence produces the obtained apparent resistivity value changed from the actual value with a constant coefficient that is based on the position and properties of the anomalous body. This phenomenon is called static shift effect (Lei et al. 2017) which is a non-inductive change of the apparent resistivity response and is identified by the lateral shift of the logarithmic apparent resistivity curves. Rugged topography can also lead to static shifts. The induced current flows laterally where it concentrates in topographic lows and spreads out in topographic highs (Fig. 3b). If the resistivity of the earth is constant, an

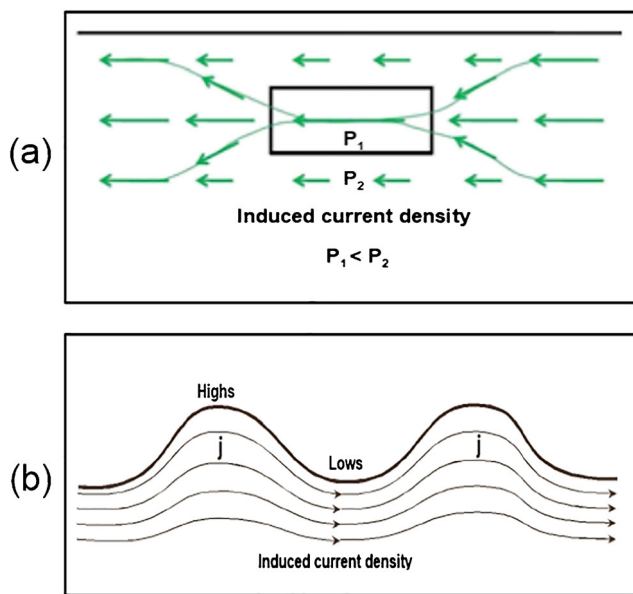


Fig. 3 (a) Current distortion due to some conductive sediments and (b) Distortion of induced current density in homogeneous earth due to topography

apparent resistivity will be higher than the true resistivity on topographic highs and vice versa (Árnason 2015).

The TDEM method only measures a secondary magnetic field and therefore does not have static shift complications (Simpson and Bahr 2005). It is relatively not affected by the near-surface inhomogeneities and/or the topography, as it is based on injection of artificial electromagnetic pluses into the earth (Irfan et al. 2010). This TDEM method loses the shallow subsurface information, because of turnoff time effect, instrumental noise, cultural or man-made noise, natural EM noise and geologic noise. Although, this loss is present, it has shown its efficiency as an integrated method with Vertical Electrical Sounding (VES) and Audio-Magnetotelluric (AMT) methods in the semiarid areas (c.f. Mohamed et al. 2002). This is attributed to its ability for penetrating thick conductive overburden overlying resistive bedrock targets (Meju et al. 1999).

The remedy of the static shift correction of the DC apparent resistivity curves and for resolving the shallow resistivity structure is joint inversion with TDEM soundings measured at the same site. Therefore, the apparent resistivity curves deduced from TDEM data were served as a leveling reference for solving the DC static shift problem in the area of study. This static shift was in similar way on all soundings in the areas of study which was in similar way on all soundings. It is caused only by local anomalous bodies due to near surface inhomogeneities; since our study area is characterized by a mild topography. This lateral shift of the apparent resistivity curves can cause errors in the inverted resistivity model because the resistivity and depths of the final model will be scaled by a scaling factor (a constant shift on log-scale) for a 1D layered model.

Inversion methods

Joint inversion of DC and TDEM data has been carried out for 32 VES using two softwares; IPI2WIN and Meju algorithm. The IPI2WIN algorithm was used to interpret DC data in terms of the layered earth model, while the Meju algorithm was used to make a joint inversion of DC and TDEM soundings and hence to correct the static shift of the DC data.

The algorithm of Meju (2005) which can be used to invert not only DC and TDEM data separately, but also make a joint inversion of DC and TDEM soundings with the correction of the static shift of DC data. The TDEM data of this sounding has been plotted in the form of log-time versus log-apparent resistivity measured at 25-time gates after the transmitter shut off. The log-electrode spacing has also been plotted versus the log-apparent resistivity for DC data at the same site. TDEM data has been transformed into true resistivity versus depth for an easy joint inversion modeling by Meju transformation (Meju 2005). DC resistivity data were also converted into true resistivity versus depth.

Results and discussions

A quantitative interpretation includes the determination of depths, thickness, and resistivity of geoelectrical layers that could be related to specific geological rock units. The lithological information obtained from well-P2, closely located nearby the study area, is correlated with the results of the inverted models, to overcome the non-uniqueness in the interpretation of resistivity curves. This well shows a succession of Quaternary sediments consisting of sands and gravels with clay intercalations and reaches a maximum depth of 250 m (Darweish 2008).

DC resistivity method is usually carried out separately by others (e.g. Attwa et al. 2014; Massoud et al. 2016). However, some others used both DC and TDEM jointly to solve the problem of static shift (e.g. Meju 2005; Sultan and Santos 2009; Leite et al. 2018). Therefore, we followed this joint inversion and shown its effectiveness in the study area. Four DC resistivity soundings were selected as examples of the inversion results (Fig. 4). The inversion models of VES (13) and VES (9) illustrate the presence of five geoelectrical layers in the agriculture and desert zones in which the data misfits were 8.89% and 5.42 %, respectively. The last layer with resistivity values 5.8 Ω .m and 30.7 Ω .m, represents probably the saturated zone of the Quaternary aquifer. Figure 5a illustrates Meju transformation for DC and TDEM data. The DC apparent resistivity curves were shifted left laterally on apparent resistivity versus depth plot without changing the shape of the curve (Fig. 5b) in order to match them with the TDEM sounding curves by scaling factor values ranging between 1.14 and 1.5. Other selective examples of the DC-TDEM joint

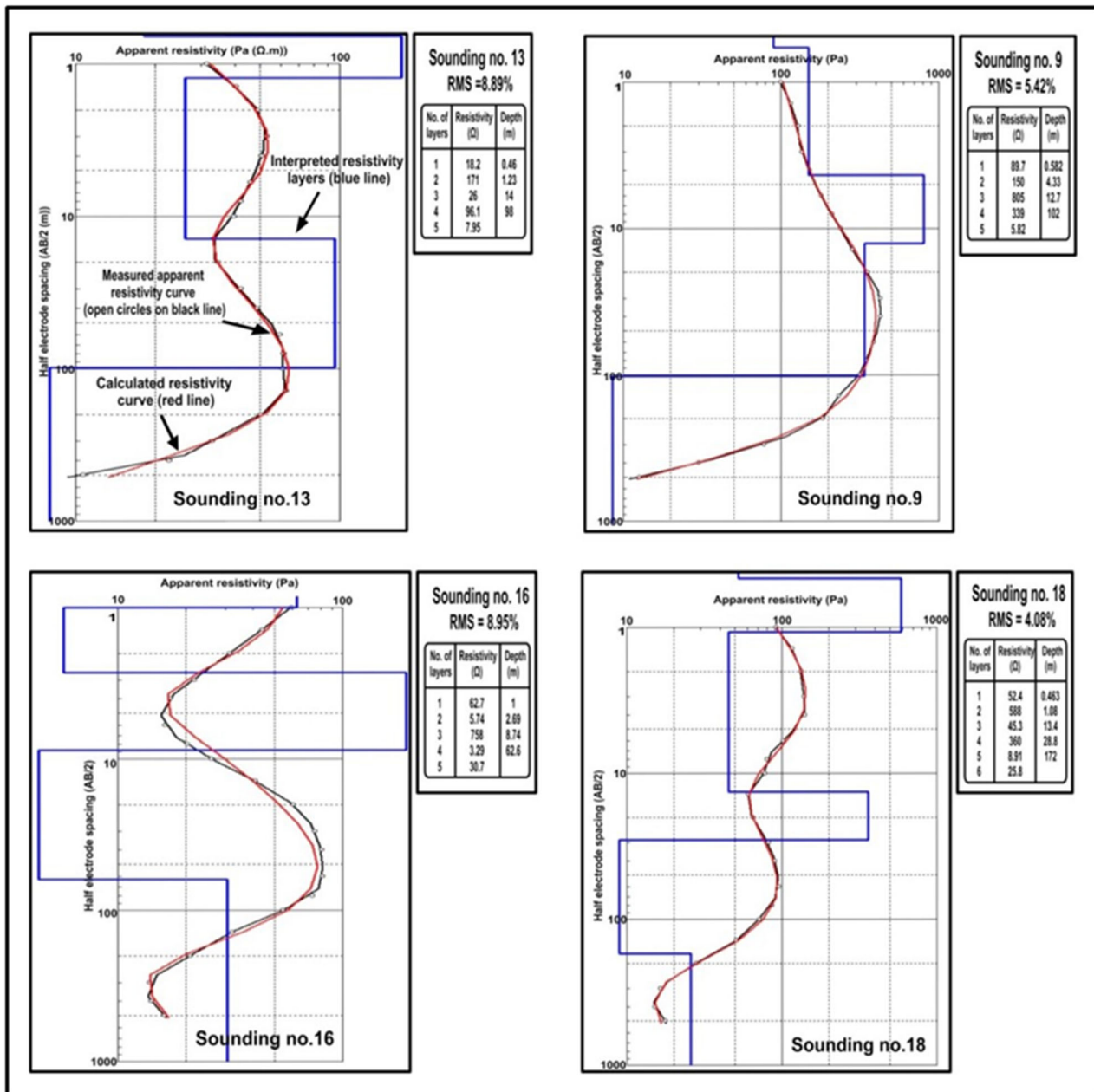


Fig. 4 Examples of Interpreted models of DC sounding curves for stations number 13, 9, 16 and 18

inversion are given in Figs. 6 and 7 for different stations. In these figures, the output resistivity model from the IPI2WIN software is used as an initial model for the DC-TDEM joint inversion. Then, some changes have been done to the initial model for the inverse modeling.

In order to obtain a regional picture of the subsurface layer distribution in the study area, the joint inversion output of the DC and TDEM data (thirty-two vertical electrical soundings) were integrated in comparison with the available borehole data to construct two geo-electrical cross-sections (A–A' and B–B') in the western part of the study area in the south-north direction (Fig. 8b,c). These sections reflect four main

subsurface layers where the lateral and vertical variations of the model parameters (true resistivity and thicknesses), are provided.

From the lithological well (P2-well) and the geological map, it is indicated that the major lithological facies are sands, gravels, and clays. These lithological facies belong to Quaternary deposits as follow;

Zone A represents a detrital surface of topsoil layer (red color) and is characterized by resistivity values ranging from 18 Ω .m (VES-13, cross-section A–A', Fig. 8b) to 228.5 Ω .m (VES-31). This topsoil contains organic

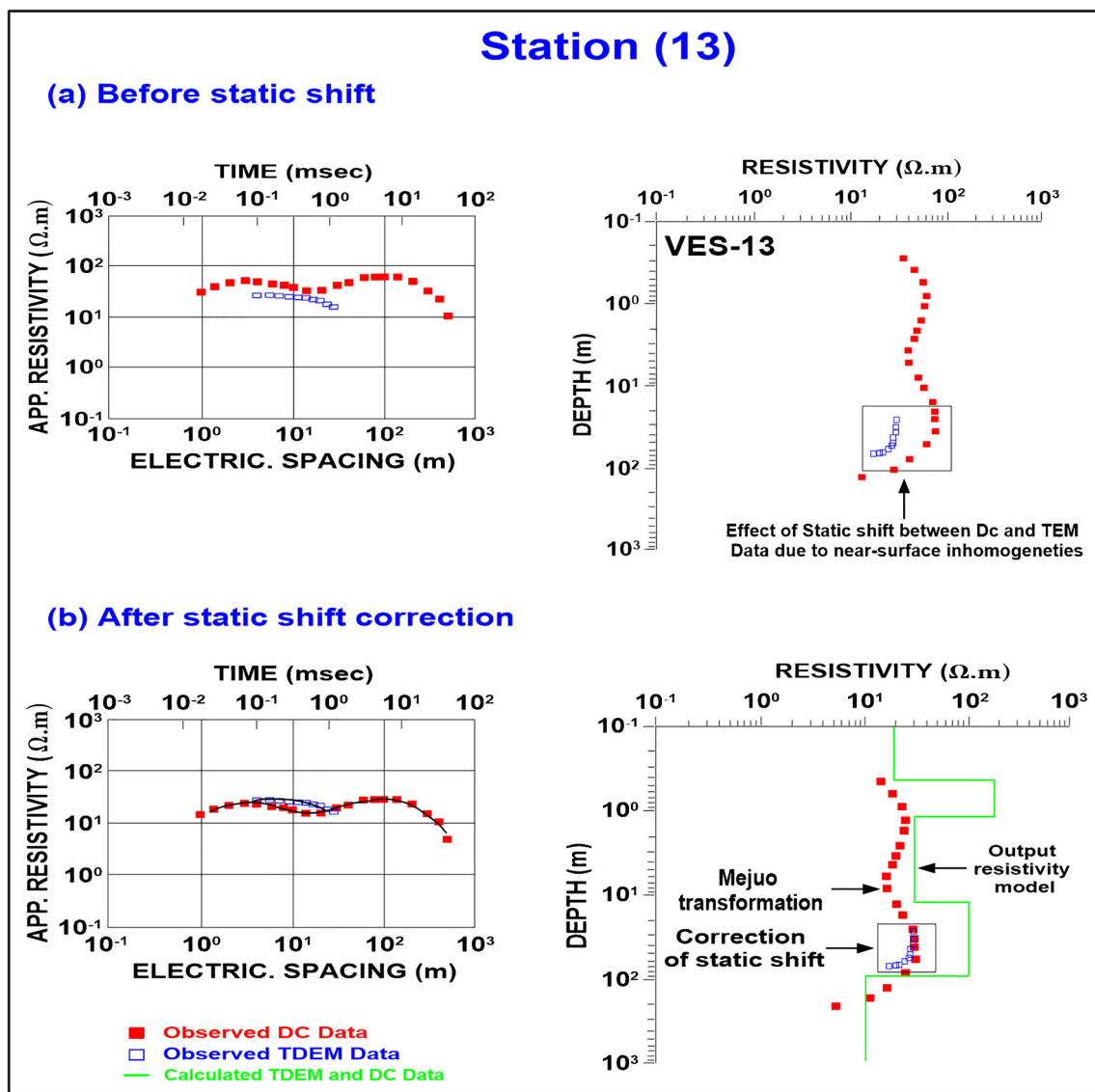


Fig. 5 Results of the DC/TDEM joint inversion in the study area at station (13) before static (a) and after static shift problem (b), and the interpreted geological models on the right side of figure (b). The DC data (red points) and TDEM data (blue points) are adjusted with the inverted curve (green curve)

matter, sand and gravel sediments with increasing or decreasing amounts of water saturation due to weather action and irrigation in this cultivated zone. So, it displays a different range of resistivity values that reflect its heterogeneity and soil water content.

Zone B consists of sand and clay lenses with calcareous materials (yellow color) and is characterized by resistivity values ranging between 5.2 $\Omega.m$ (VES-30) and 48.5 $\Omega.m$ (VES-8). The highest resistivity values within this layer are in the south direction (especially, below VES_{es} 15, 9, 8, 31 and 32) and this may be attributed to the presence of gravels (709 $\Omega.m$, at VES-15 to 1721 $\Omega.m$, at VES-31). The thickness of this layer decreases toward the north.

Zone C is characterized by high resistivity values ranging from 15.5 $\Omega.m$ (VES-14) to 336 $\Omega.m$ (VES-7) and its

thickness ranges from 27.24 m below VES-1 (Fig. 8c) to 126.5 m below VES-14 (Fig. 8b). The high resistivity values may be attributed to the presence of sands and gravels (chart reuse color), especially below VES_{es} (30, 10, 11, 9 and 7). These values decrease laterally toward the north, especially below VES-14, due to the presence of a thick clay zone having a thickness of 59 m (green color). This resistivity value decreases vertically below VES-14 as a result of the presence of sand and some clay intercalations (Fig. 8b).

Zone D represents a water-bearing formation. It consists probably of saturated sands (clayey sands) that may represent the free Quaternary aquifer (ice blue color). It exhibits resistivity values ranging from 3.5 $\Omega.m$ (VES-31) to 21.7 $\Omega.m$ (VES-10). The average depth to the top of

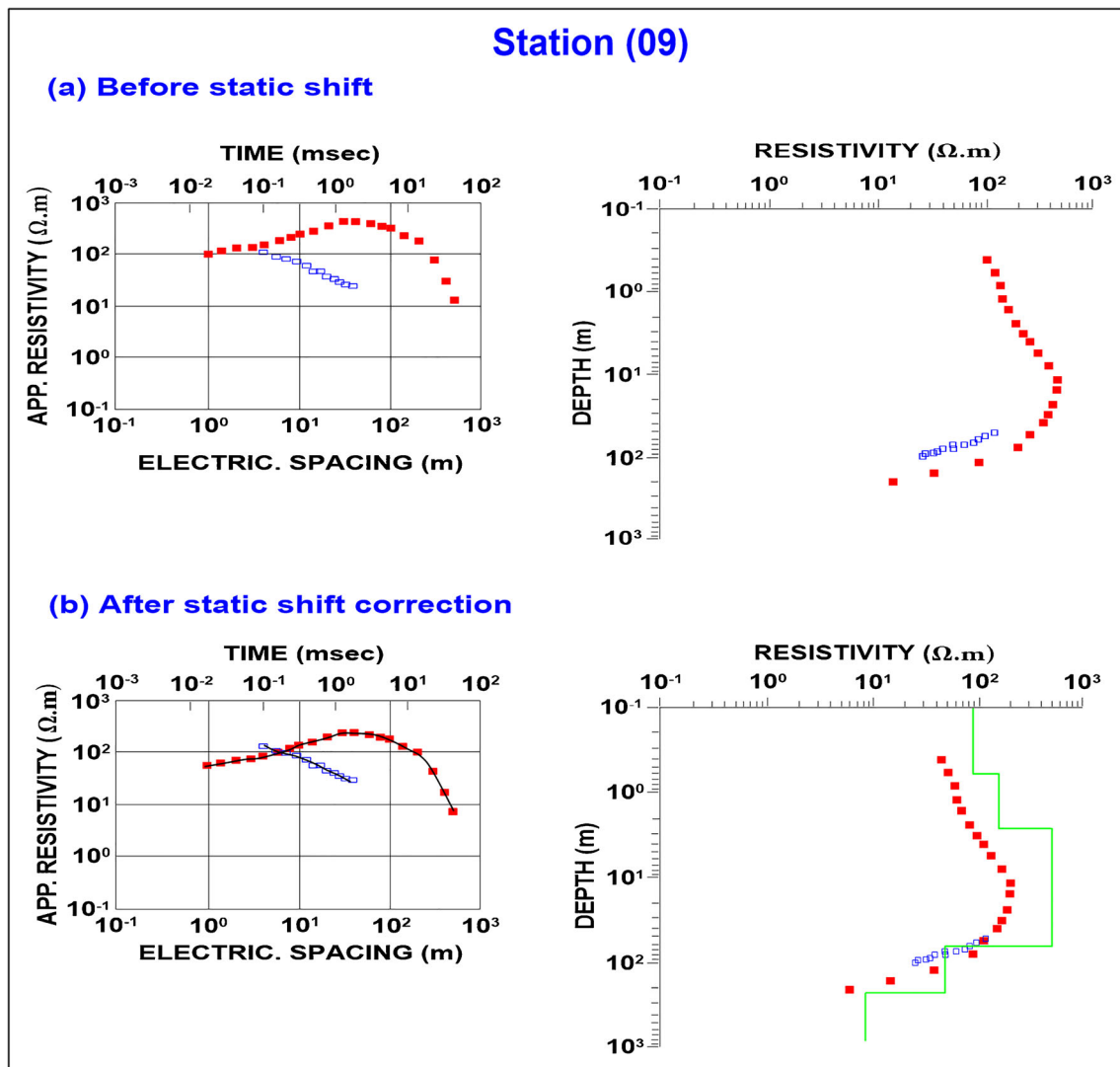


Fig. 6 Results of the DC/TDEM joint inversion in the study area at station (9) before static (a) and after static shift problem (b), and the interpreted geological models on the right side of figure (b)

this layer ranges between 51 m and 139 m along main profiles which agrees with the information encountered from the borehole (P2-well) where its top depth begins from 60 m. Its lower boundary is not reached and, consequently, its thickness is not defined, as the used maximum distance of AB/2 was 500 m. The lowest resistivity values at depths from 50.57 m (VES-1, Fig. 8c) to 138.80 m (VES-14, Fig. 8b) could be attributed to water saturated horizon that rises southward due to faulting effect causing relative topographic highs. The south part of the study area is highly affected by two normal step faults along these sections trending mainly E–W and NE–SW directions; forming horst structures. These faults are easily identified due to abrupt changes in the values of resistivity on both sides of the interpreted fault plane with a significant amount of throw. These faults may control and play an important role in groundwater movement

and interaction between the deeper and shallow aquifers (Gad et al. 2015; Khalaf and Gad 2015). So, upward water leakage along this fault plane (VES-1 and VES-30) might control the quality and hydrological parameters of the groundwater. Moreover, the horst shape below VES-1 (Fig. 8c) plays an important role as a barrier fault, which might prevent the lateral movement of groundwater inside this aquifer.

Conclusions

The Joint inversions of DC and TDEM data have been carried out for 32 VES stations in the study area. The IPI2WIN algorithm is used to interpret DC data in terms of the layered earth model while the Meju algorithm is used to make a joint inversion of DC and

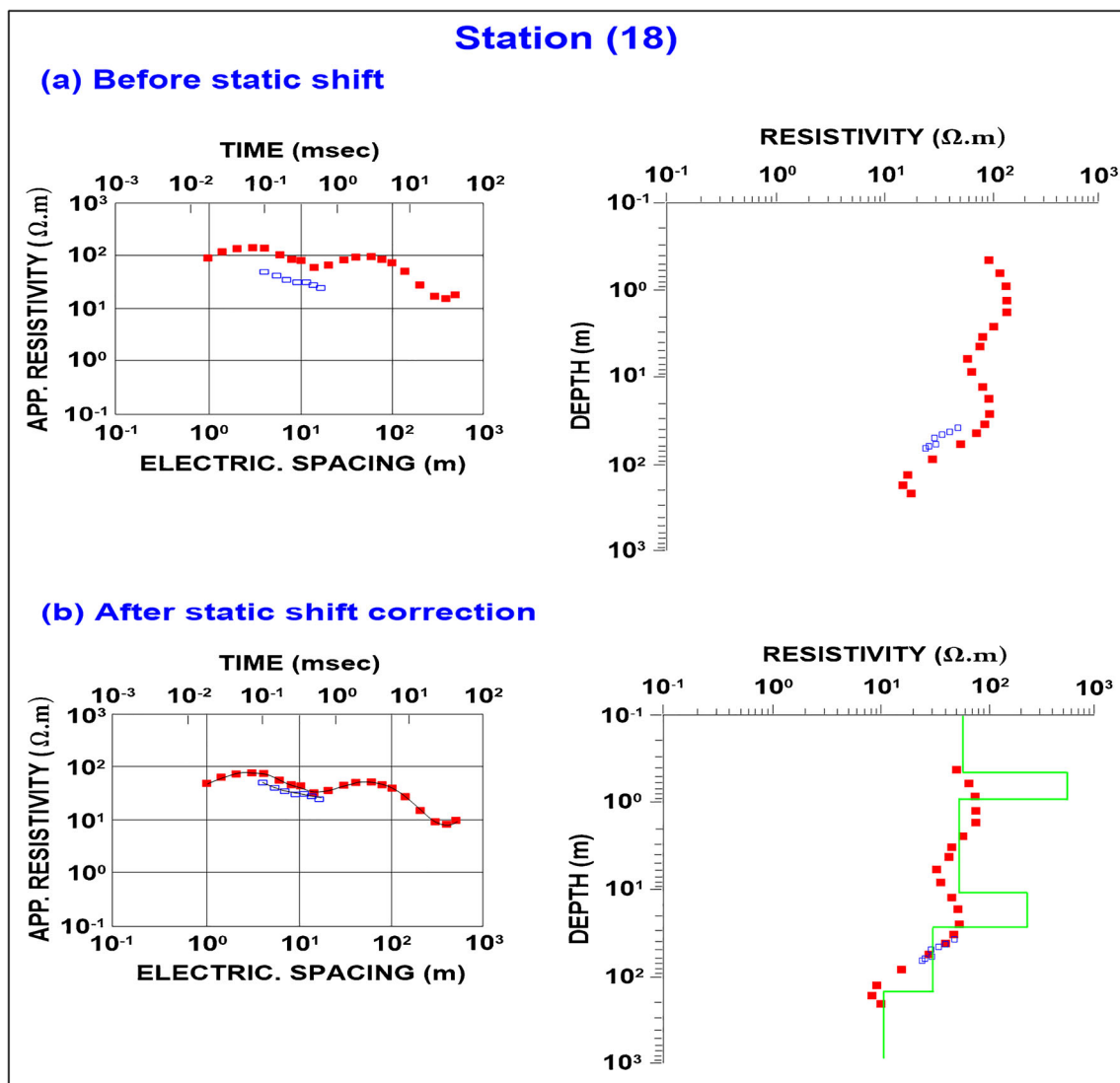


Fig. 7 Results of the DC/TDEM joint inversion in the study area at station (18) before static **(a)** and after static shift problem **(b)**, and the interpreted geological models on the right side of figure **(b)**

TEM data and to correct the static shift of DC data, caused by a local anomalous body (near-surface inhomogeneities) that severely impede the interpretation of the geoelectric data.

The output results indicated that the DC-TDEM Joint inversion technique is an effective tool for supporting the data interpretation. The joint inversion technique has been used in conjunction with DC data of the same site in order to correct the static shift by scaling factor values ranging between 1.14 and 1.5. Five geoelectrical layers are identified. Among them, the last layer has resistivity values between $5.8 \Omega.m$ and $30.7 \Omega.m$ which represent the saturated zone of the Quaternary aquifer. These five layers were tied with the available lithology of P2-well.

Two geo-electrical cross-sections are constructed to represent lateral and vertical variations of these subsurface layers and their parameters (true resistivity and thicknesses). The first layer which is represented by aeriated detrital surface deposits such as Quaternary sand sheets, gravels, clays, and alluvial deposits has

resistivity values ($18\text{--}228.5 \Omega.m$). The second layer consists of sand and clay lenses with calcareous materials ($5.2\text{--}48.5 \Omega.m$) and shows decreasing thickness northward.

The third layer is composed of dry sands and gravels with some clay intercalations in the northern part of the study area and has resistivity values ($15.5\text{--}336 \Omega.m$) with thickness ranges from 27.24 to 126.5 m. The last layer with resistivity values from $3.5 \Omega.m$ to $21.7 \Omega.m$ is reached at a depth ranging between 51 and 139 m. It consists probably of saturated sand that may represent the water-bearing formations of Quaternary aquifer. Finally, there is a good agreement between our depth results (51 m–139 m) and the information recorded from borehole (P2-well; 60 m–80 m) to the top depth level of Quaternary aquifer zone for the last layer.

The southern part of the study area is dissected by step normal faults forming horst structures that control upward water leakage and interconnection between the shallow and deep Quaternary free aquifers. They may act as a

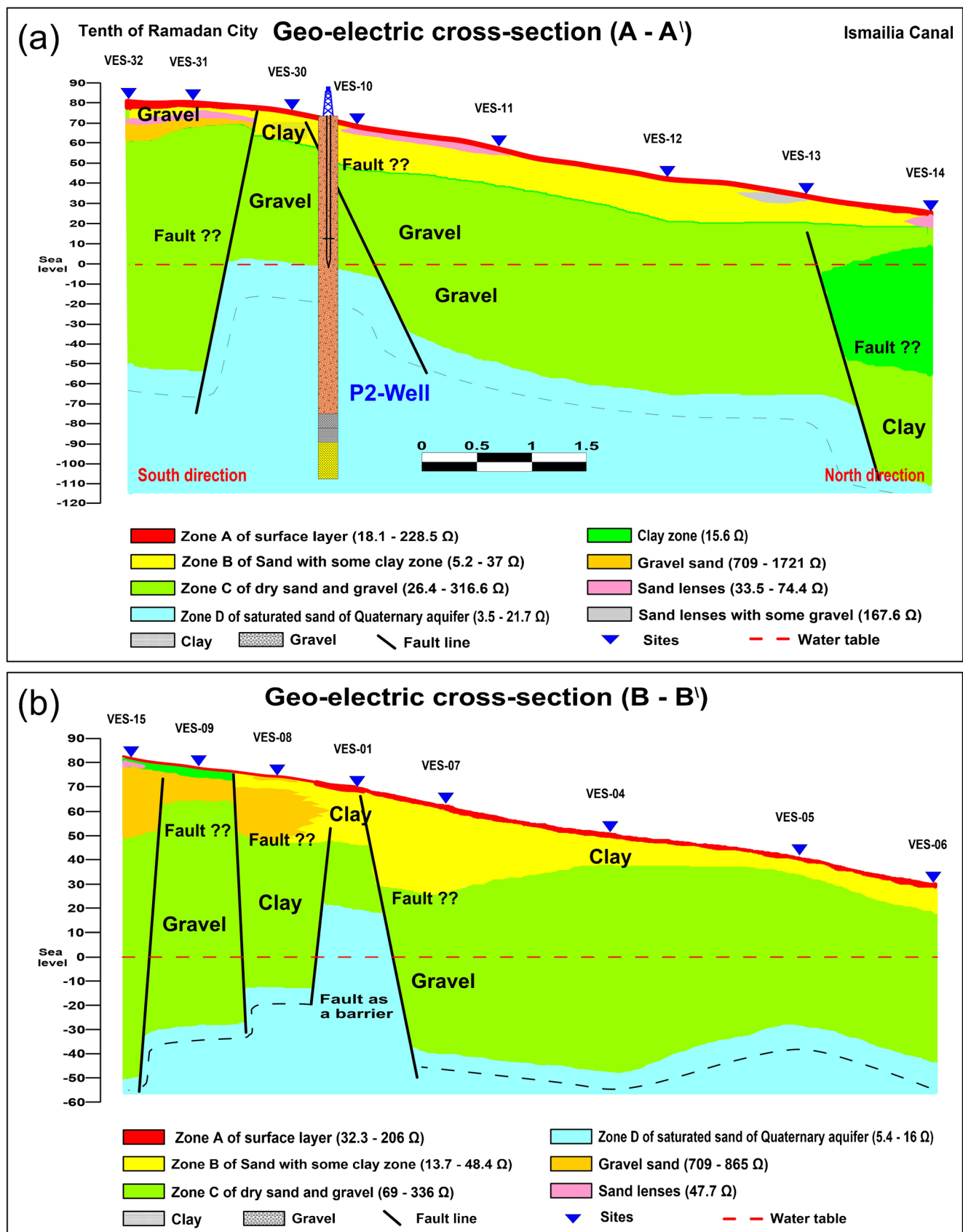


Fig. 8 Geoelectric cross-section constructed from DC/TDEM joint inversion results along two longitudinal profiles (A-A') with correlation of P2-well and profile (B-B') that are located in the west of the study area

barrier and prevent the lateral movement of groundwater inside this aquifer. The obtained results gave more reliable subsurface geological model as we could be able to overcome the limitations of DC and TDEM surveys in such noisy area of study. Also, we recommended to use magnetotelluric survey to study the deeper aquifers (Miocene aquifer) jointly with the TDEM technique.

Acknowledgements We would like to acknowledge Prof. Bulent Tezkan, Institute of Geophysics and Meteorology, Cologne University, Germany, for reviewing the manuscript and his valuable comments.

Declarations

Conflict of interest The author(s) declare that they have no competing interests.

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